SIMATS ENGINEERING

CO2 - ASSESSMENT TOOL 2

Scenario-Based Questions

COURSE NAME: Deep Learning **COURSE CODE: MLA0408**

COURSE OUTCOME COVERED

CO 2: Students will be able to apply supervised learning algorithms such as linear regression, classification models, decision trees, neural networks, support vector machines, and ensemble methods to solve real-world prediction and classification problems. They will gain the ability to select appropriate models, train them using datasets, and evaluate their performance. Students will also understand the strengths and limitations of different supervised learning approaches. (BTL-4)

CO Weightage: 15%

Assessment Tool Weightage: 40%

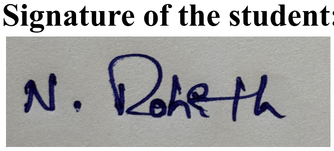
Duration: 90 minutes

Marks: 25

Code of Conduct:

I **NALIPI REDDY ROHITH/192425115**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Scenario-Based Questions

CO-AT2

QUESTIONS: 05 **Total Marks:25**

1. **Unsupervised Training of Neural Networks, Auto Encoders**
   A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

2. **Auto Encoders, Representation Learning**
   A retail company wants to reduce the dimensionality of customer purchase data before performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

3. **Width and Depth of Neural Networks, Activation Functions**
   A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

4. **Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks**
   A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

5. **Deep Learning Applications, Representation Learning**

   A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Scenario-Based Questions

Q1. Unsupervised Training of Neural Networks, Auto Encoders

Question:
A hospital wants to identify abnormal ECG patterns from thousands of unlabeled records. Which deep learning technique from the syllabus would you recommend? Justify your choice and explain the workflow.

Answer:

I would recommend using an **Autoencoder**, which is an unsupervised neural network used for anomaly detection. Since the ECG data is unlabeled, the autoencoder learns the normal patterns by compressing and reconstructing the input data. During testing, abnormal ECG signals produce a higher reconstruction error because they differ from the learned normal patterns. This helps identify irregular heart conditions without requiring labeled data.

Workflow:

1. Collect unlabeled ECG records.
2. Train the autoencoder using normal ECG patterns.
3. Compress and reconstruct the ECG signals.
4. Calculate the reconstruction error.
5. If the error exceeds a threshold, classify the ECG as abnormal.

Conclusion: Autoencoders efficiently detect abnormal ECG patterns from large unlabeled datasets.

Q2. Auto Encoders, Representation Learning

Question:
A retail company wants to reduce the dimensionality of customer purchase data before

performing customer segmentation. Explain how an autoencoder can be used to solve this problem.

Answer:

An **Autoencoder** can reduce the number of features by learning a compact representation of customer purchase data. The encoder compresses the original high-dimensional data into a smaller latent representation while preserving important information. The decoder reconstructs the original data from this compressed form. The compressed features are then used for customer segmentation using clustering algorithms such as K-Means.

Advantages:

- Removes redundant information.
- Reduces storage and computation.
- Improves clustering performance.
- Preserves important customer behavior patterns.

Conclusion: Autoencoders perform effective dimensionality reduction and improve customer segmentation accuracy.

Q3. Width and Depth of Neural Networks, Activation Functions

Question:
A facial recognition system performs poorly when trained using a shallow neural network. Recommend architectural changes involving network depth and activation functions.

Answer:

A shallow neural network cannot learn complex facial features effectively. Increasing the **depth** by adding more hidden layers enables the network to learn low-level, mid-level, and high-level facial features. Using the **ReLU activation function** improves gradient flow and speeds up training compared to sigmoid or tanh. Batch Normalization and Dropout can also be added to improve training stability and reduce overfitting.

Recommended Changes:

- Increase the number of hidden layers.
- Use ReLU activation.
- Apply Batch Normalization.
- Add Dropout layers.
- Train with a larger dataset.

Conclusion: A deeper network with ReLU activation improves facial recognition accuracy and generalization.

Q4. Restricted Boltzmann Machines (RBMs), Unsupervised Training of Neural Networks

Question:
A streaming service wants to recommend movies to users based on their viewing history without using labeled data. Explain how RBMs can be applied to build the recommendation engine.

Answer:

A **Restricted Boltzmann Machine (RBM)** is an unsupervised neural network that learns hidden patterns from user viewing history. Users and movies are represented as visible units, while hidden units capture user preferences. After training, the RBM predicts movies that users are likely to enjoy based on learned patterns. Similar users receive similar recommendations without requiring labeled data.

Workflow:

1. Collect user viewing history.
2. Train the RBM using viewing data.
3. Learn hidden user preferences.
4. Predict missing movie ratings.
5. Recommend the highest-rated movies.

Conclusion: RBMs provide personalized movie recommendations by learning user preferences from unlabeled viewing data.

Q5. Deep Learning Applications, Representation Learning

Question:
A manufacturing company wants to detect defective products from sensor readings collected every second. Design a deep learning solution using concepts from representation learning and neural networks.

Answer:

A deep learning solution can use an **Autoencoder** or a **Deep Neural Network** to analyze sensor readings. The model first learns meaningful representations of normal sensor data. During operation, new sensor readings are passed through the trained network. If the reconstruction error or prediction error is high, the product is identified as defective. This enables automatic quality inspection in real time.

Workflow:

1. Collect sensor readings from products.
2. Preprocess and normalize the data.
3. Train the neural network using normal data.
4. Extract meaningful features using representation learning.
5. Detect defective products based on high reconstruction error.
6. Generate alerts for defective products.

Conclusion: Representation learning with neural networks enables fast and accurate detection of defective products while reducing manual inspection.